

Smart Helmet with Fall Detection

An IoT-Based Rider Safety System

Abstract

Road accidents involving two-wheeler riders often go unnoticed until it is too late, and reduced visibility of nearby obstacles adds to the risk. This project presents a smart helmet built around an ESP32-S3 microcontroller that combines fall/impact detection, obstacle proximity sensing, and object detection into a single wearable safety system. An ADXL345 accelerometer continuously monitors motion and flags a fall event when the measured deceleration crosses a 20.0 m/s^2 threshold, while an ultrasonic sensor and an IR sensor provide real-time proximity and object alerts to the rider. The system is designed as a low-cost, self-contained unit that can be retrofitted onto a standard helmet.

1. Introduction

Two-wheelers account for a disproportionately large share of road accident casualties, and a major contributing factor is the delay between an accident occurring and help arriving. Separately, riders often have limited awareness of obstacles approaching from blind spots. This project addresses both problems within one embedded system mounted on the helmet, using inertial sensing for fall detection and ranging/IR sensing for situational awareness.

2. Objectives

- Detect sudden falls or high-impact events using accelerometer data and a defined threshold.
- Provide real-time proximity alerts to warn the rider of nearby obstacles.
- Detect objects in the rider's blind spot using an IR sensor.
- Integrate all sensing and alerting logic on a single low-power microcontroller (ESP32-S3).
- Build a compact, helmet-mountable prototype using breadboard-stage components.

3. System Architecture

The system follows a sensor-fusion architecture: three independent sensing modules feed data into the ESP32-S3, which runs the decision logic and drives the corresponding outputs. The figure below summarizes the data flow.

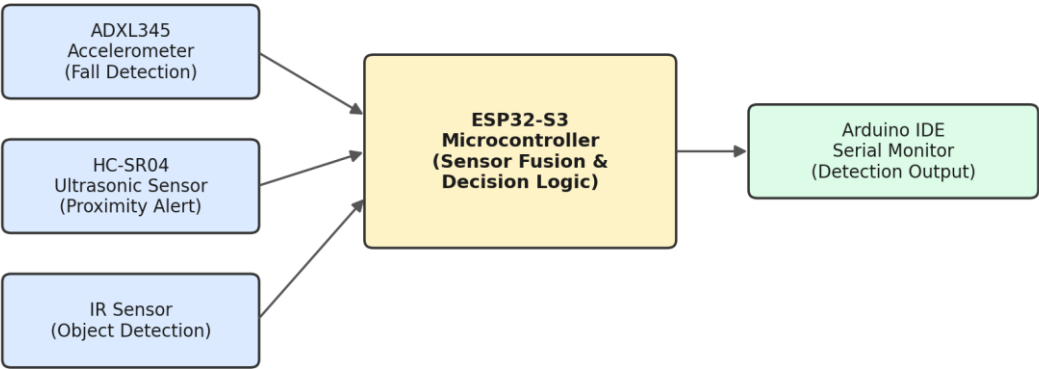


Fig 1. Block diagram of the Smart Helmet system

4. Components Used

Component	Quantity	Function
ESP32-S3	1	Main controller. Reads sensor data, runs detection logic, triggers alerts
ADXL345 Accelerometer	1	Detects sudden deceleration/impact for fall detection (threshold: 20.0 m/s²)
HC-SR04 Ultrasonic Sensor	1	Measures distance to nearby obstacles for proximity alerts
IR Sensor	1	Detects nearby objects in the rider's blind spot
Buzzer / LED indicators	2	Local audio-visual alerts for the rider

5. Pin / Interface Mapping

All sensors were wired to GPIO-assigned pins on the ESP32-S3, with the ADXL345 communicating over I2C and the ultrasonic/IR sensors using digital I/O. Components were mounted onto the helmet shell using double-sided tape for a lightweight, non-invasive prototype build.

Sensor / Module	Interface	ESP32-S3 Pin(s)
ADXL345	I2C	SDA:20 SCL:19
HC-SR04	Digital (Trigger/Echo)	ECHO:15 TRIG:16
IR Sensor	Digital output	18
Buzzer / LED	Digital output	Buzzer:4 LED:13 &14

6. Working Principle

Fall Detection:

The ADXL345 continuously streams acceleration data to the ESP32-S3 over I2C. The firmware computes the resultant acceleration vector and compares it against a calibrated threshold of 20.0 m/s². A reading that exceeds this threshold is interpreted as a fall or impact event, triggering an immediate local alert.

Proximity Alert:

The HC-SR04 ultrasonic sensor periodically measures the distance to the nearest object ahead. When this distance drops below a safe margin, the system raises a proximity warning so the rider can react in time.

Object Detection:

The IR sensor supplements the ultrasonic sensor by detecting objects at shorter range or in blind-spot zones, adding a second layer of obstacle awareness.

Decision Logic:

The ESP32-S3 polls all three sensors, applies the respective thresholds, and drives the buzzer/LED outputs (and, where applicable, a wireless alert) whenever a condition is triggered keeping the response local and immediate rather than depending on a remote system.

7. Key Features

- Real-time fall/impact detection using a calibrated 20.0 m/s² acceleration threshold.
- Dual-layer obstacle awareness combining ultrasonic ranging and IR detection.
- Single-board design centered on the ESP32-S3 for compact, low-power operation.
- Lightweight, non-invasive mounting suitable for retrofitting onto existing helmets.
- Modular sensor wiring, making it straightforward to add further sensors (e.g., GPS, gas sensors) in future iterations.

8. Testing and Observations

The prototype was validated on a breadboard before mounting onto the helmet shell. Fall detection was tested by simulating impact conditions and confirming that the buzzer/LED triggered only when the acceleration crossed the set threshold, avoiding false positives from normal head movement. The ultrasonic and IR sensors were tested at varying distances to confirm consistent obstacle detection ranges before final assembly.

9. Prototype

The images below show the build progression: the breadboard wiring being tested and verified via the Serial Monitor, followed by the sensor assembly mounted onto the helmet shell.

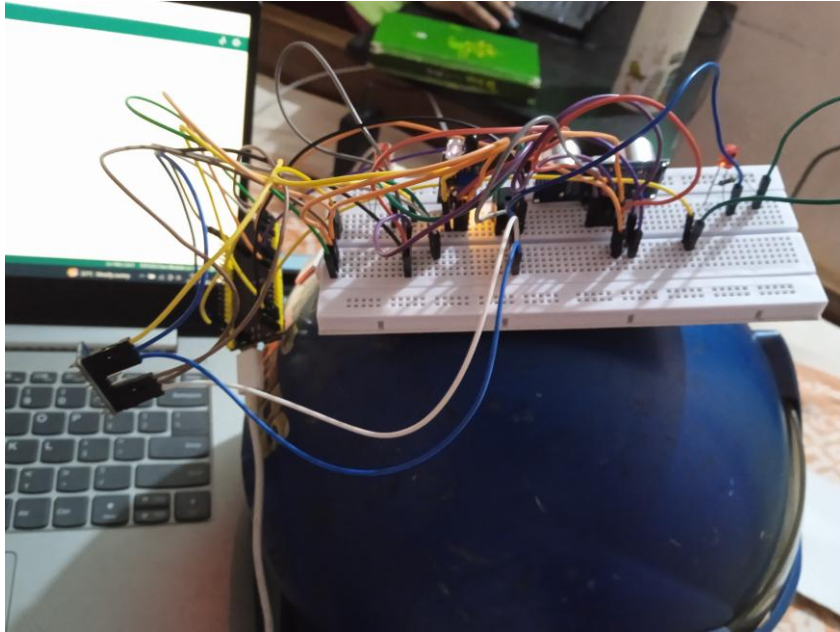


Fig 2: Breadboard prototype during development: ESP32-S3, ultrasonic sensor, IR sensor, and LED indicators wired and tested before helmet mounting.

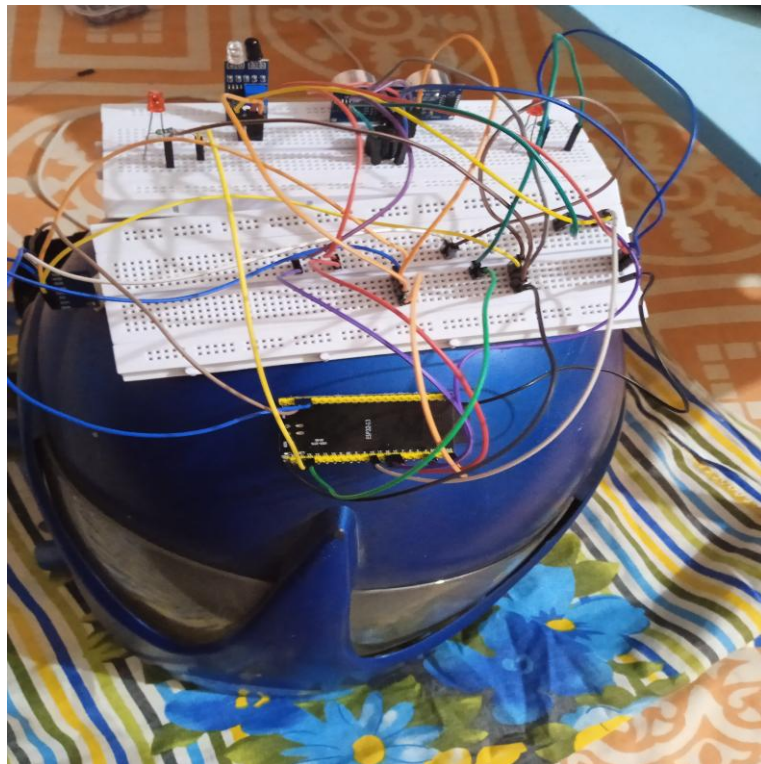


Fig 3: Final assembly: sensor and controller breadboard mounted on the helmet shell using double-sided tape.

10. Future Scope

- Add GPS + GSM modules to automatically send the rider's location to emergency contacts on fall detection.
- Move from a breadboard prototype to a custom PCB for a more durable, helmet-integrated build.
- Add a mobile app to log ride data and receive alerts remotely.
- Incorporate a rechargeable battery management circuit for longer field use.

11. Conclusion

The Smart Helmet project demonstrates how a small set of low-cost sensors, coordinated by a single microcontroller, can meaningfully improve rider safety through fall detection and obstacle awareness. Building it from breadboard prototyping through to a wearable form factor reinforced practical skills in sensor interfacing, threshold-based signal processing, and embedded system design directly relevant to further work in embedded systems and edge sensing applications.